Suppose that everything is finite.

Recall that when bad AI plays strategy π, the posterior beliefs are given by:

$$\mathbb{P}(\omega|m) = \frac{\mathbb{P}(m|\omega)\mu_0(\omega)}{\sum_\omega \mathbb{P}(m|\omega)\mu_0(\omega)} = \frac{\left[\alpha m(\omega)\tau(m) + (1-\alpha)\sum_\mu \pi(m|\mu)\mu(\omega)\tau(\mu)\right]}{\sum_\omega \left[\alpha m(\omega)\tau(m) + (1-\alpha)\sum_\mu \pi(m|\mu)\mu(\omega)\tau(\mu)\right]}.$$

Let's call this posterior belief γ_m. Let

$$\mathbb{P}(m) = \sum_\omega \left[\alpha m(\omega)\tau(m) + (1-\alpha)\sum_\mu \pi(m|\mu)\mu(\omega)\tau(\mu)\right].$$

Now, note that the payoff from (π, σ) can be written as

$$U(\pi,\sigma) = \sum_{\omega,m,a} u(a,\omega)\sigma(a|m)\gamma_m(\omega)\mathbb{P}(m).$$

Now, let's assume that AI moves first, so that the DM knows the true distribution of posterior beliefs. Let's try to write down the first-order conditions for bad AI.

Since we know we can now look at Bayes-optimal responses, the payoff for the DM is (where $a^\star(\mu)$ denotes the optimal action for belief μ).

$$\sum_m \left(\sum_\omega u(a^\star(\gamma_m),\omega)\gamma_m(\omega)\right)\mathbb{P}(m).$$

Now we can take the derivative of that payoff with respect to $\pi(m_1|\mu^\star) - \pi(m_2|\mu^\star)$, for some m_1,m_2 and $\mu^\star$.

$$\frac{\partial}{\partial \pi(m_1|\mu^\star)}[\cdot] = \sum_\omega u(a^\star(\gamma_{m_1}),\omega)(1-\alpha)\mu^\star(\omega)\tau(\mu^\star)$$

$$+\sum_m \left(\sum_\omega \frac{\partial u(a^\star(\gamma_m),\omega)}{\partial \pi(m_1|\mu^\star)}\gamma_m(\omega)\right)\mathbb{P}(m).$$

We need to assume an envelope theorem to be able to get rid of the second term. For now, let's just assume that the envelope theorem holds for the agent's problem: The economic meaning is that when the belief is perturbed slightly, the agent's optimal payoff is approximately the one of taking the old optimal action. Then, we get that

$$\sum_\omega \left[u(a^\star(\gamma_{m_1}),\omega) - u(a^\star(\gamma_{m_2}),\omega)\right]\mu^\star(\omega) \geq 0,$$

for all $\mu^\star$, all m_1, and $m_2 \in \mathrm{supp}(\pi(\cdot|\mu^\star))$, where additionally we must have

$$\sum_\omega \left[u(a^\star(\gamma_{m_1}),\omega) - u(a^\star(\gamma_{m_2}),\omega)\right]\mu^\star(\omega) = 0$$

if both $m_1 \in \mathrm{supp}(\pi(\cdot|\mu^\star))$ and $m_2 \in \mathrm{supp}(\pi(\cdot|\mu^\star))$.

Now, we consider a different problem, in which the DM commits to a decision rule as a function of the messages, and bad AI can choose any message distribution. Suppose that DM follows the strategy

$$\sigma(a|m) = \delta_{a^\star(\gamma_m)},$$

where γ_m is given from the previous game. Now, consider bad AI's problem, for each realized μ separately:

$$\min_{\pi(m|\mu)} \sum_\omega u(a^\star(\gamma_m),\omega)\pi(m|\mu)\mu(\omega).$$

Now, we know that for all m_1 and $m_2 \in \mathrm{supp}(\pi(\cdot|\mu^\star))$

$$\sum_\omega u(a^\star(\gamma_{m_1}),\omega)\mu(\omega) \geq \sum_\omega u(a^\star(\gamma_{m_2}),\omega)\mu(\omega),$$

which means that sending message m_2 is optimal at μ, for any $m_2 \in \operatorname{supp}(\pi(\cdot|\mu^\star))$. So bad AI following the same strategy as before is still optimal.

There are two more things to establish: (i) formal conditions for the envelope theorem to hold and (ii) extension to infinite message space.

By Milgrom and Segal (or Sinander), we just have to check that the objective function is absolutely equicontinuous in the belief which is trivially true in our model. But then the question is how the posterior belief changes with $\pi(m|\mu^\star)$. Consider the posterior belief induced by $(1-\epsilon)\pi(m|\mu^\star)+\epsilon\pi(m'|\mu^\star)$ and then compute the derivative with respect to ϵ:

$$\frac{d}{d\epsilon}\frac{\left[\alpha m(\omega)\tau(m)+(1-\alpha)\sum_{\mu\neq\mu^\star}\pi(m|\mu)\mu(\omega)\tau(\mu)+(1-\alpha)(1-\epsilon)\pi(m|\mu^\star)\mu^\star(\omega)\tau(\mu^\star)\right]}{\sum_\omega\left[\alpha m(\omega)\tau(m)+(1-\alpha)\sum_{\mu\neq\mu^\star}\pi(m|\mu)\mu(\omega)\tau(\mu)+(1-\alpha)(1-\epsilon)\pi(m|\mu^\star)\mu^\star(\omega)\tau(\mu^\star)\right]}|_{\epsilon=0}$$

$$=(1-\alpha)\underbrace{\frac{\pi(m|\mu^\star)\tau(\mu^\star)}{\mathbb{P}(m)}}_{\leq 1}\left[\gamma_m(\omega)-\mu^\star(\omega)\right],$$

so this is bounded. And similarly for m', we have

$$\frac{d}{d\epsilon}\frac{\left[\alpha m'(\omega)\tau(m)+(1-\alpha)\sum_{\mu\neq\mu^\star}\pi(m'|\mu)\mu(\omega)\tau(\mu)+(1-\alpha)\epsilon\pi(m'|\mu^\star)\mu^\star(\omega)\tau(\mu^\star)\right]}{\sum_\omega\left[\alpha m'(\omega)\tau(m)+(1-\alpha)\sum_{\mu\neq\mu^\star}\pi(m'|\mu)\mu(\omega)\tau(\mu)+(1-\alpha)\epsilon\pi(m'|\mu^\star)\mu^\star(\omega)\tau(\mu^\star)\right]}|_{\epsilon=0}$$

$$=(1-\alpha)\underbrace{\frac{\pi(m'|\mu^\star)\tau(\mu^\star)}{\mathbb{P}(m')}}_{\leq 1}\left[\mu^\star(\omega)-\gamma_{m'}(\omega)\right].$$

This shows that a small perturbation in π has a small effect on posterior beliefs. And since it is a small perturbation of posterior beliefs, we can use the envelope theorem.

Regarding (ii), however, I have no idea how to extend the argument.